

Muhammad Ibrahim

(+92) 309-424-2532 · ibrahimanwarulhaq1994@gmail.com · LinkedIn · Portfolio

TECHNICAL SKILLS

- **AI & LLM Systems:** GPT-4, LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Function Calling, Conversation Memory, OpenRouter
- **Backend Development:** FastAPI, Node.js, RESTful APIs, Redis, WebSockets, JWT Authentication, Docker
- **Frontend Engineering:** React.js, Next.js, TypeScript, Responsive Web Design
- **Languages:** Python, JavaScript, SQL (PostgreSQL, MySQL)
- **Data Systems & Tools:** ETL Pipelines, Data Validation, Git, Tableau, SSIS

EXPERIENCE

IBEX

Software Engineer

Lahore, Pakistan

Nov 2024 – Present

- Embedded with an enterprise client, enhancing their AI customer service agent on Sierra's platform using both code (Agent SDK) and no-code (Agent Studio) tooling.
- Built and maintained full-stack features for IBEX's internally developed AI Agent Builder Platform (AMP) using React.js and FastAPI, enabling internal teams to configure, test, and deploy intelligent voice and chat agents.
- Designed and implemented Redis-based session management, conversation history tracking, and function-calling integrations to enable context-aware, low-latency LLM interactions on AMP.
- Collaborated with AI teams on RAG-based architectures to connect domain-specific enterprise knowledge bases to agent pipelines.
- Developed and maintained internal Node.js web applications used by IBEX call center agents, supporting day-to-day operational workflows.

PROJECTS

AI Agent Builder Platform (AMP)

[Platform Link](#)

Tech Stack: Python, FastAPI, React.js, OpenAI API, Redis, WebSockets, Docker

- Developed full-stack web features for a configurable AI agent platform supporting native voice and chat interfaces, used by internal prompt engineers and operations teams.
- Implemented custom conversational workflows integrating advanced LLM function calling and RAG knowledge pipelines to enable domain-aware agent responses.
- Containerized services using Docker to ensure consistent, reproducible environments across development and production deployments.

MeetBot – AI-Driven Meeting Assistant

[GitHub](#) · [Live Demo](#)

Tech Stack: Next.js, TypeScript, FastAPI, RAG, OpenRouter API, Notion API

- Built an end-to-end AI meeting assistant that ingests raw transcripts and converts them into a structured, searchable knowledge base using RAG and LLM summarization.
- Developed a responsive Next.js and TypeScript frontend paired with a high-performance FastAPI backend capable of handling high-throughput transcript ingestion.
- Implemented speaker-aware chunking and vectorized retrieval with LLM-driven summarization via OpenRouter API, improving response relevance for meeting queries.
- Engineered asynchronous background processing jobs and a Notion API integration to automatically export meeting summaries and action items post-meeting.

EDUCATION

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Science in Computer Science (CGPA: 3.42 / 4.0)

Oct 2020 – Jun 2024

- **Final Year Project:** Suspicious Behavior Detection using CCTV – Awarded **3rd Position** at the Department Industrial Open House for excellence in real-time computer vision modeling.